



PLENUMNET · ENERGY ATTESTATION

SPECIFICATION

Energy Attestation Certificate

TM-2026-042 Rev.2 · EAC/1

DOCUMENT

|        |                                                                                                                                                                                                  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title  | <div>Energy Attestation Certificate (EAC)</div> <div>A signed, blockchain-witnessed proof of how much electrical energy this PlenumNET node saved during the measurement window.</div>           |
| Issuer | <div>PlenumNET node · tdns:hmodal-demo:01</div> <div>The PlenumNET node that produced and signed this attestation.</div>                                                                         |
| Mode   | <div>compute-throughput-proxy</div> <div>hardware-watts = real CPU power counters (RAPL). compute-throughput-proxy = modeled from real measured CPU compute time when RAPL is not exposed.</div> |
| Demand | <div>auto</div> <div>Workload pattern during the measurement window (idle / steady / burst / auto sine sweep).</div>                                                                             |

TIMESTAMP · UTC-GROUNDED · ATTOSECOND PRECISION

ATTOSECONDS SINCE UTC EPOCH · 1970-01-01T00:00:00Z

1 777 428 699 226 081 926 728 003 226 as

= 2026-04-29T02:11:39.226Z (UTC, ISO 8601)

Plain English: the exact wall-clock instant this certificate was issued, expressed as a single integer count of attoseconds ( $10^{-18}$  s) since the UTC Unix epoch. True attosecond precision end-to-end: the UTC epoch was anchored ONCE to the framework's attosecond tick at process start; every subsequent certificate advances purely along the framework's monotonic attosecond walk — no millisecond rounding, no synthetic clamps, no per-call wall-clock reads.

trit (Rep-C): 233112321 213121333 113111213 123232132 111332231 231133213 123

|                            |                                                                                                                                                                                                                |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware clock used        | <div>NO — pure framework derivation</div> <div>The OS / wall-clock was NOT consulted. Time is derived from a deterministic tick walk on the integer ring <math>\mathbb{Z}_{\{D_\alpha\}}</math>.</div>         |
| Tick counter (decimal)     | <div>405</div> <div>Number of HPTP framework ticks since boot. Each tick is an exact rational of attoseconds.</div>                                                                                            |
| Tick (trit)                | <div>112223</div> <div>The same tick counter expressed in Rep-C bijective base-3 (digits {1,2,3} only).</div>                                                                                                  |
| Clock tier                 | <div>0 (0 = pure derivation)</div> <div>0 means the timestamp comes from pure framework math; higher tiers mean increasing reliance on hardware clocks.</div>                                                  |
| Chain index at seal (trit) | <div>123</div> <div>Which sealed sample on the WSS attestation chain this certificate snapshots.</div>                                                                                                         |
| Chain tag at seal (trit)   | <div>111321112 223211213 122132122 131221132 233133233 332223112 ...</div> <div>The 385-bit running TL-Sponge tag at the moment of seal — any reorder or substitution of WSS samples changes this value.</div> |

► Show audit-purity derivation (rational form, walk modulus, factorisation)

MEASUREMENT · POWER & ENERGY SAVINGS

|                                 |                                                                                                                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Window (ms)                     | <div>3800</div> <div>Length of the measurement window, in milliseconds.</div>                                                                                                               |
| Measured power                  | <div>1000 mW</div> <div>Average power draw of this node during the window, in milliwatts.</div>                                                                                             |
| Baseline power                  | <div>5000 mW</div> <div>What the node would have drawn if it had run a continuous-on (always full-throttle) workload during the same window.</div>                                          |
| Power saved                     | <div>4000 mW</div> <div>baseline – measured = how many milliwatts of continuous draw the HModal duty-cycle eliminated.</div>                                                                |
| Savings ratio                   | <div>4000 / 5000 (≈ 80.00%)</div> <div>Fraction of baseline power eliminated, kept as an exact integer ratio (no float on the wire).</div>                                                  |
| Savings ratio (theoretical)     | <div>143 / 192 (74.48%)</div> <div>Closed-form best case for the canonical <math>\alpha=91/36</math>, <math>\beta=91/3</math>, duty 1:4 schedule.</div>                                     |
| Cumulative energy consumed (μJ) | <div>3 813 000</div> <div>Total energy this node actually consumed since the WSS session opened, in microjoules.</div>                                                                      |
| Cumulative energy SAVED (μJ)    | <div>15 252 000</div> <div>Total electrical energy this node DID NOT spend by running HModal duty-cycled instead of continuous-on, since the WSS session opened. Integer microjoules.</div> |
| Cumulative energy SAVED (trit)  | <div>231123212 133213</div> <div>The same saved-energy count expressed in Rep-C bijective base-3.</div>                                                                                     |

CRYPTOGRAPHIC INTEGRITY · REP-C BIJECTIVE BASE-3 (TRIT-NATIVE)

|                                |                                                                                                                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cipher                         | <div>TL-Sponge-385 duplex (Phase Encryption v3)</div> <div>Post-quantum 385-bit duplex sponge — the same primitive that signs the cert also seals every WSS sample.</div>                                         |
| Session ID (trit)              | <div>321211112 322321212 233333121 333113311 211312332 332132211 123313123 221223 222 1221331</div> <div>Unique identifier for this WSS measurement session, generated trit-native (no hex stage).</div>          |
| Session-key fingerprint (trit) | <div>123231232 121332212 111112322 131112331 113212331 133231313 ...</div> <div>TIS-27 fingerprint of the per-session sponge key — proves the cert was sealed by THIS session's keystream.</div>                  |
| Chain seed (trit)              | <div>323213132 132113131 231311212 221322313 312232232 323223312 112311221 321111 333 321222211 ...</div> <div>Initial sponge state before the first sample was sealed — the anchor of the integrity chain.</div> |
| Chain index (decimal)          | <div>18</div> <div>Sequence number of the sealed sample this cert binds to.</div>                                                                                                                                 |
| Chain index (trit)             | <div>123</div>                                                                                                                                                                                                    |

|                             |                                                                                                                                                                                                                                   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chain tag · 385-bit (trit)  | 332322211 313112133 123212232 313221233 221322131 221223321 223121111 312123213 223333121 ...<br>Running 385-bit TL-Sponge tag after sealing this sample. ANY gap, reorder, or substitution in the WSS stream changes this value. |
| Cipher payload (trit, head) | 322313333 311123231 311123212 121221321 323312322 322222222 211133213 131322131 111223122 ...                                                                                                                                     |
| TIS-27 doc hash (trit)      | 121313232 211231233 322113111 123313333 123123333 133232321 ...<br>Hash of the canonical EAC JSON document, computed with the trit-native TIS-27 sponge.                                                                          |
| TIS-27 · Milesian glyphs    | πζβφβδηξμρζλζρλϣζωαβφλτζκ<br>The same hash rendered in Milesian Greek numerals — visual cross-check for transcription errors.                                                                                                     |

SIGNATURE

|                        |                                                                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Variant                | TL-DSA-87<br>Post-quantum 87-byte digital signature — Salvi Framework's ternary lattice DSA.                                                                                                               |
| Public key hash (trit) | 331332132 121112213 211312123 121132221 333211313 332211331 ...<br>TIS-27 fingerprint of the signer's public key. Compare to the node's published key roster.                                              |
| Signature (trit, head) | 212321312 123313323 211313332 311133222 113122121 313122133 112133321 131113331 233121331 ...<br>The TL-DSA-87 signature over the canonical EAC JSON. Rep-C trit form (full hex copy in the audit footer). |

HEDERA HCS WITNESS · BLOCKCHAIN NON-REPUDIATION

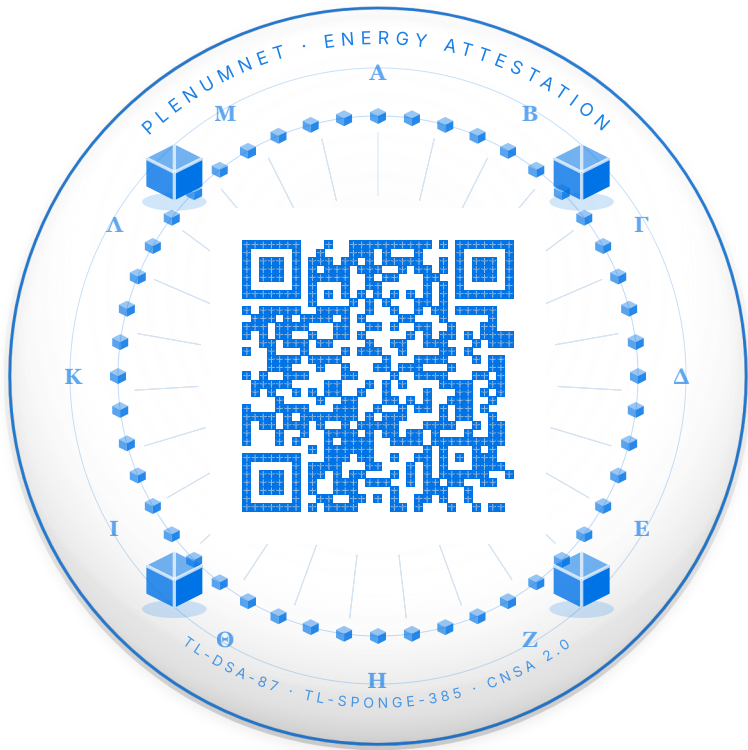
|           |                                                                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Status    | Not configured on this node<br>Hedera HCS witnessing is available in the framework but no operator credentials are present on this node. The signed cert is still valid; it just lacks the optional blockchain anchor. |
| To enable | Set HEDERA_ACCOUNT_ID and HEDERA_PRIVATE_KEY                                                                                                                                                                           |

CALENDAR STAMP · 42-SYSTEM MULTI-CIVILIZATIONAL SYNC

Plain English: the same instant rendered into twelve historical calendar systems, derived purely from the Julian Day Number — provides a civilization-independent anchor for the cert's "when".

|                      |                                                   |                    |                                                    |
|----------------------|---------------------------------------------------|--------------------|----------------------------------------------------|
| GREGORIAN (UTC)      | 2026-04-29T02:15:04.104Z                          | JULIAN DAY NUMBER  | —                                                  |
| MAYAN LONG COUNT     | —                                                 | HEBREW             | 12 Iyar 5786 AM                                    |
| ISLAMIC HIJRI        | 12 Dhu al-Qidah 1447 AH                           | CHINESE SEXAGENARY | —                                                  |
| VEDIC KALI YUGA      | Kali Yuga Year 5,127 of 432,000 (1.1868% elapsed) | PERSIAN SOLAR      | 10 Ordibehesht 1405 SH                             |
| ETHIOPIAN GE'EZ      | 21 Miazia 2018 (Ethiopian)                        | COPTIC             | 21 Parmouti 1742 AM (Coptic)                       |
| EGYPTIAN CIVIL       | Year 4806, Akhet (Inundation), Month 4, Day 29    | 13-MOON HARMONIC   | Lunar Moon, Day 1 (Dali), Year 2026 [Cycle 30,026] |
| BYZANTINE ANNO MUNDI | 16 Aprilios, Anno Mundi 7,534, In diction 4       |                    |                                                    |

salvi-core/ancient-calendar-sync (42 systems, JDN-anchored)



- Scannable · 3-D isometric crystal lattice (ISO/IEC 18004)
- 36-cube mandala · 4 corner anchors · 27 sponge rays

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AUDIT INTEROP · LEGACY HEX COPIES (DO NOT CONSUME)
tis27_hash_hex: d54b51749bba5d214382eab46e6b7f6c
chain_tag_hex: 1f994d4413916388cf123ab7d7215643...
public_key_hash: 16eb7a1083a9450bd2532a0c24f462ae...
```





